

Weekly Diary – Week 13

Internship Organization: Vikaspedia

Duration: 14/04/2026 – 20/04/2026

Objective of the Week

The primary objective of this week was to implement the fingerprinting functionality within the system, establish proper integration between the frontend and backend components, and thoroughly test the tracking mechanism. The focus was on ensuring accurate data flow, reliable session handling, and correct identification of users across multiple visits.

Date – 14 April

On this day, API integration was implemented to transmit the generated fingerprint ID along with user event data from the frontend to the backend server. Special care was taken to ensure that the data transmission process was consistent and reliable, allowing all user interactions to be captured and sent without loss or delay.

Date – 15 April

Backend logic was developed to receive the incoming fingerprint data and store it efficiently in the database using MySQL. Guest session records were successfully created, ensuring that each unique fingerprint was properly recorded along with its associated information.

Date – 16 April

Functionality was implemented to update existing session records based on the fingerprint ID. This allowed the system to recognize returning users and update relevant fields such as the last_seen timestamp. As a result, user activity across multiple sessions could be tracked accurately without creating duplicate entries.

Date – 17 April

End-to-end testing of the complete system was performed. This included verifying fingerprint generation, data transmission, storage, and tracking processes. The testing ensured that the system operated correctly and that no duplication issues occurred while identifying users.

Date – 18 April

Additional testing was conducted to evaluate the system's functionality and accessibility across different devices. The goal was to ensure that the fingerprinting mechanism worked consistently regardless of device type or environment, confirming its reliability in real-world scenarios.

Date – 19 April

On this day, the final project presentation was prepared, summarizing the system design, implementation process, and key features. The completed project was also uploaded to platforms such as GitHub and Google Drive for submission and future reference.